| Candidate Name | Centre Number | Candidate Number |
|---|---|---|
| | | |

**International General Certificate of Secondary Education**

**CAMBRIDGE INTERNATIONAL EXAMINATIONS**

# CHEMISTRY 0620/2

PAPER 2

**OCTOBER/NOVEMBER SESSION 2002**

1 hour

Candidates answer on the question paper.
No additional materials are required.

**Time** 1 hour

**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets [ ] at the end of each question or part question.

You may use a calculator.

A copy of the Periodic Table is printed on page 16.

| FOR EXAMINER'S USE | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| TOTAL | |

**This question paper consists of 16 printed pages.**

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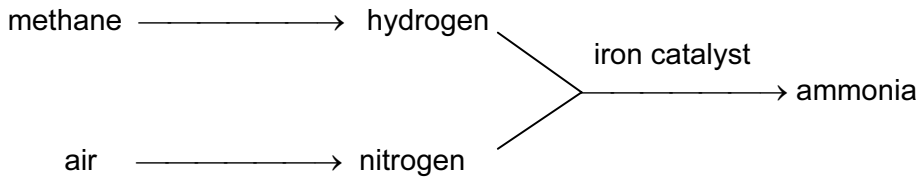
[Turn over

For Examiner's Use

**1** Ammonia, $NH_3$, is synthesised by the following route.




**(a) (i)** To which group of organic compounds does methane, $CH_4$, belong?

Put a ring around the correct answer.

**alkane** **alcohol** **alkene** **carboxylic acid**

[1]

**(ii)** Draw the formula for methane, showing all atoms and bonds.

[1]

**(iii)** State the most likely source of methane.

..... [1]

**(b) (i)** State the percentage of nitrogen in clean air.

..... [1]

**(ii)** Name another non-metal that is in the same Period as nitrogen.

..... [1]

**(c)** Ammonia is made by heating hydrogen with nitrogen in the presence of a catalyst.

**(i)** What is the purpose of the catalyst?

..... [1]

**(ii)** What happens to the rate of a reaction when the temperature is increased?

..... [1]

For Examiner's Use

**(d)** **(i)** Complete the following equation which shows the synthesis of ammonia from hydrogen and nitrogen.

$$3H_2 \quad + \quad N_2 \rightleftharpoons \ldots NH_3$$ [1]

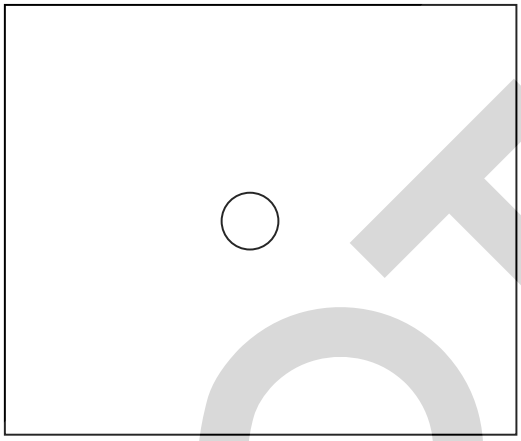
**(ii)** What does the sign $\rightleftharpoons$ mean?

..... [1]

**(e)** The ammonia formed in the reaction is liquefied.

Complete the diagram below to show the arrangement of the molecules in liquid ammonia.

○ represents a single molecule of ammonia.



[2]

**(f)** How would you test for ammonia in the laboratory?

test .....

result ..... [2]

**(g)** Ammonia is used to make fertilizers.

**(i)** Why are fertilizers used in agriculture?

..... [1]

**(ii)** Some fertilizers contain ammonium sulphate.

Complete the word equation to show how ammonium sulphate is formed.

ammonia + ..... ..... ⟶ ammonium sulphate

[1]

ROOT 19

For Examiner's Use

**2** When rain water trickles through rocks, it dissolves some of the minerals present.

This water, which is bottled for drinking, is called mineral water.

The table shows the ions present in a litre of mineral water.

| name of ion | formula of ion | mass of ion present in one litre of water/milligrams |
|---|---|---|
| calcium | $Ca^{2+}$ | 10 |
| chloride | $Cl^{-}$ | 8 |
| hydrogencarbonate | $HCO_3^{-}$ | 64 |
| sodium | $Na^{+}$ | 8 |
| sulphate | $SO_4^{2-}$ | 7 |

**(a)** What do you understand by the term *ion*?

..... [1]

**(b)** Which positive ion has the greatest concentration in this sample of water?

..... [1]

**(c)** Complete the following equation to show how a calcium ion is formed from a calcium atom.

$$Ca \longrightarrow Ca^{2+} + \ldots e^{-}$$

[1]

**(d)** When this sample of mineral water is evaporated to dryness, various compounds are formed. One of these compounds is calcium chloride.

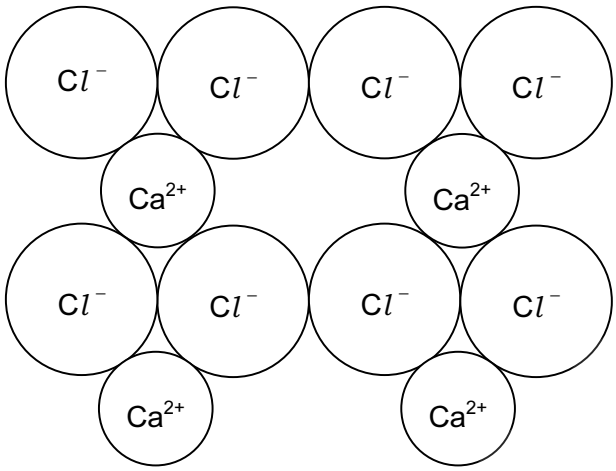
Suggest the name of **two** other compounds which could be formed.

compound 1 .....

compound 2 ..... [2]

For Examiner's Use

**(e)** Part of the structure of calcium chloride is shown below.




Use this diagram to work out the simplest formula for calcium chloride.

formula ..... [1]

**(f)** Complete the following table to show the electrical conductivity of calcium and calcium chloride in the solid and liquid states.

Put a ✓ if the substance conducts.

Put a ✗ if the substance does not conduct.

| substance | state | electrical conductivity |
|---|---|---|
| calcium | solid | |
| calcium | liquid | |
| calcium chloride | solid | |
| calcium chloride | liquid | |

[2]

**(g)** A sample of water was contaminated with clay, which is insoluble in water.

Explain with the help of a labelled diagram, how you would separate the clay from the water.

[3]

**3** Fluorine, chlorine, bromine and iodine are halogens.

**(a)** Complete the table by filling in the blank spaces.

| halogen | colour | melting point /°C | boiling point /°C | state at room temperature |
|---|---|---|---|---|
| fluorine | yellow | -220 | -188 | |
| chlorine | | -101 | -35 | gas |
| bromine | reddish-brown | -7 | +59 | |
| iodine | | +114 | | solid |

[4]

**(b)** Predict the boiling point of iodine.

[1]

**(c)** When chlorine is bubbled through a solution of potassium bromide, the solution turns orange - red.

When iodine is mixed with potassium bromide, no colour change occurs.

**(i)** Write a word equation for the reaction between chlorine and potassium bromide.

[2]

**(ii)** Put the elements bromine, chlorine and iodine in order of reactivity.

most reactive → [ ]

[ ]

least reactive → [ ]

[1]

**(d)** State a use of chlorine.

..... [1]

For Examiner's Use

**(e)** In the presence of sunlight, chlorine reacts with methane.

Hydrogen chloride gas, H — C*l*, is given off during this reaction.

State the type of bonding in a hydrogen chloride molecule.

Put a ring around the correct answer.

**covalent** **ionic** **metallic** **weak**

[1]

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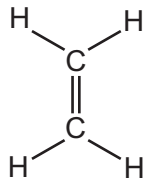
For Examiner's Use

**4** Some organic compounds found in ripe fruits are shown below.



**A**

$CH_3CO_2H$

**B**

$CH_3CH_2CH_2CO_2H$

**C**

$CH_3CH_2OH$

**D**

$CH_3CH_2CHO$

**E**

**(a)** What do you understand by the term *organic compound*?

.....

..... [1]

**(b)** Which **two** of the compounds belong to the same homologous series?

compound ..... and compound ..... [1]

**(c)** Which **one** of these compounds is an unsaturated hydrocarbon?

..... [1]

**(d)** Which **one** of these compounds is an alcohol?

..... [1]

**(e)** Which **one** of these compounds can be formed directly by cracking the paraffin fraction from petroleum?

..... [1]

**(f)** Compound **D** burns readily.

**(i)** Burning is an exothermic reaction.

Explain the meaning of the term *exothermic*.

..... [1]

**(ii)** State the products formed when **D** burns in excess air.

..... [2]

**(iii)** Name the carbon compound formed when **D** undergoes incomplete combustion.

..... [1]

**(g)** Write down the molecular formula of compound **C**.

..... [1]

**(h)** Calculate the relative molecular mass of compound **C**.

..... [1]

**(i)** Many fruits contain a variety of different coloured compounds.

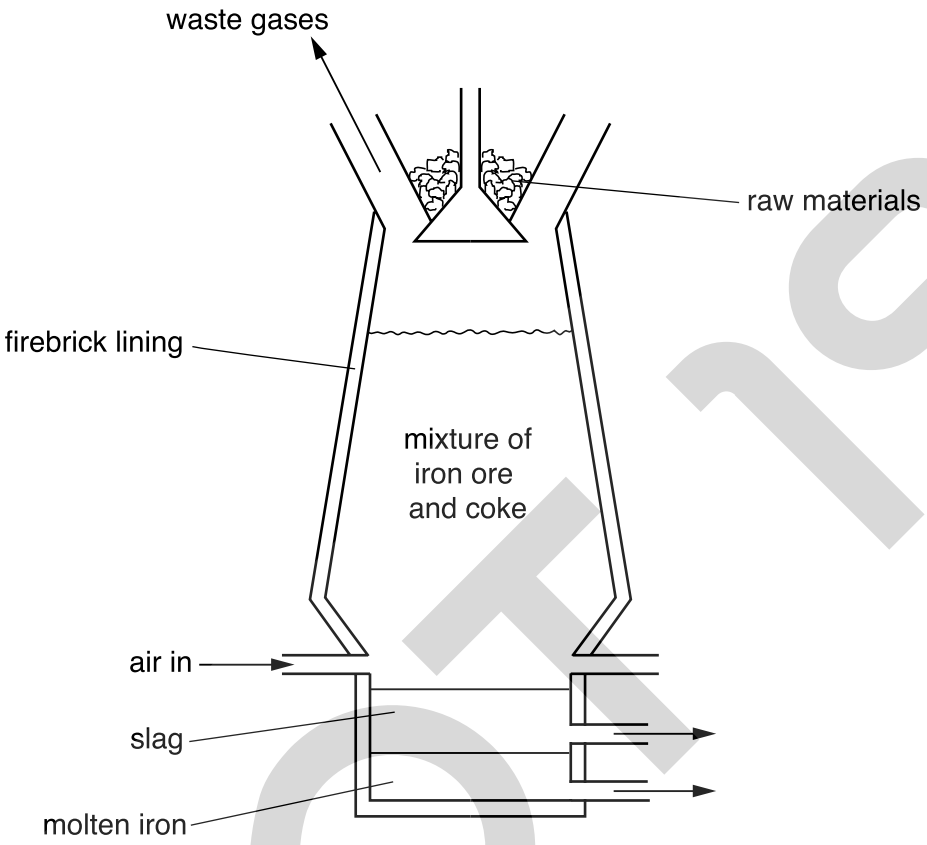
What separation technique can you use to separate these different coloured compounds?

..... [1]

For Examiner's Use

**5** Iron is extracted from the ore, haematite.

The iron ore is put in a blast furnace with coke and a current of air is blown through the heated mixture.




**(a)** What do you understand by the term *ore*?

.....

..... [1]

**(b)** What other raw material needs to be added to the blast furnace?

Put a ring around the correct answer.

**cement** **limewater** **limestone** **slag**

[1]

For Examiner's Use

**(c)** Near the bottom of the furnace, iron(III) oxide is reduced by carbon.

$$Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$$

**(i)** Write a word equation for this reaction.

[1]

**(ii)** Explain what is meant by the term *reduction*.

.....

..... [1]

**(d)** The table shows the composition of the waste gases leaving the blast furnace.

| gas | percentage of gas in the mixture |
|---|---|
| carbon dioxide | 12 |
| carbon monoxide | 24 |
| hydrogen | 4 |
| nitrogen | 60 |

**(i)** The hydrogen in the waste gas is formed by the reaction of hot carbon with water vapour.

There is no water in the materials added to the top of the furnace.

Suggest where this water vapour comes from.

..... [1]

**(ii)** The reaction of hot carbon with water vapour is endothermic.

What is meant by the term *endothermic*?

..... [1]

**(e)** Iron can be converted into steel, which is more resistant to corrosion.

**(i)** Describe briefly how iron is converted into steel.

.....

.....

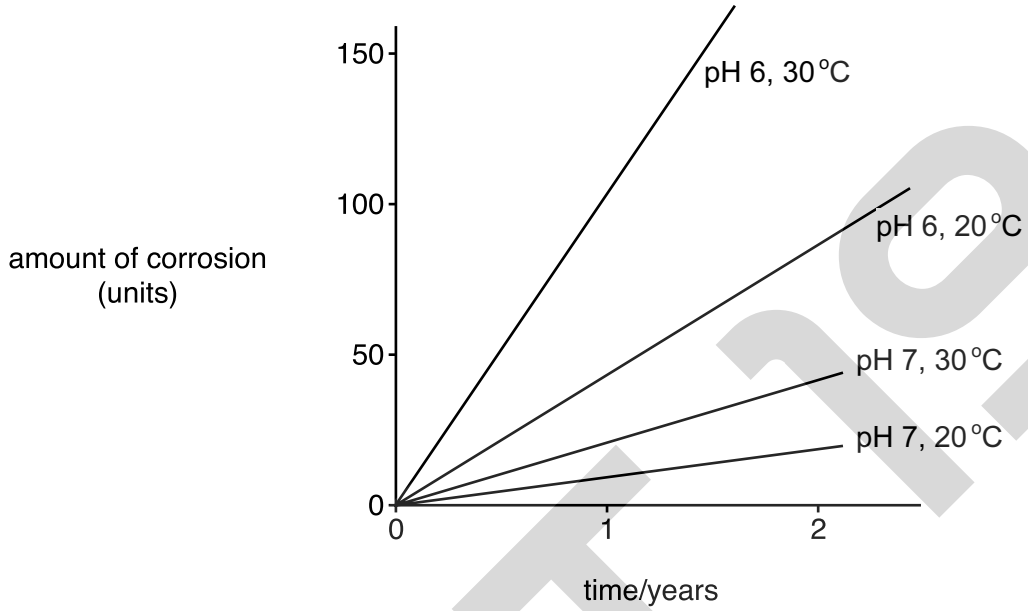
..... [2]

**(ii)** State **one** use of mild steel.

..... [1]

**(f)** In some conditions, steel corrodes more quickly than in others.

The graphs show the rate of corrosion of a particular type of steel under different controlled conditions.




**(i)** How does pH affect the rate of corrosion?

..... [1]

**(ii)** How does temperature affect the rate of corrosion?

..... [1]

Explain this in terms of moving particles.

.....

..... [2]

**(iii)** The presence of acidic gases in the air may increase the rate of corrosion.

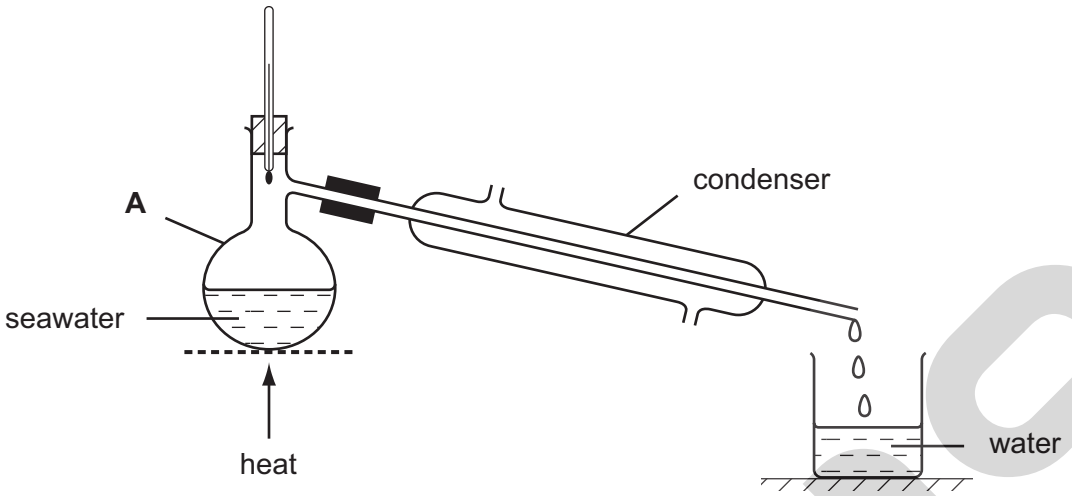
State the name and source of **one** acidic gas found in the air as a result of pollution.

name .....

source ..... [2]

For Examiner's Use

**6** A student took a sample of seawater and heated it using the apparatus shown below.




**(a)** What is the name given to the process shown in the diagram?

..... [1]

**(b)** State the name of the piece of apparatus labelled **A**.

..... [1]

**(c)** Explain the function of the condenser.

.....

.....

..... [2]

**(d)** Pure water collects in the beaker.

**(i)** State the pH of pure water.

..... [1]

**(ii)** State the boiling point of pure water.

..... [1]

**(e)** The table shows the mass of various compounds obtained when 1 litre of seawater is evaporated.

| compound | formula | mass of solid present / g |
|---|---|---|
| sodium chloride | $NaCl$ | 28.0 |
| | $MgCl_2$ | 8.0 |
| magnesium sulphate | $MgSO_4$ | 6.0 |
| calcium sulphate | $CaSO_4$ | 2.0 |
| potassium chloride | $KCl$ | 1.0 |
| calcium carbonate | $CaCO_3$ | |
| potassium bromide | $KBr$ | |
| | | total mass = 45.0 |

**(i)** How many grams of magnesium sulphate are present in 180 g of solid left by evaporation of seawater?

[1]

**(ii)** Which compound in the table reacts with acids to release carbon dioxide?

..... [1]

**(iii)** State the name of the compound which has the formula $MgCl_2$.

..... [1]

**(iv)** Calcium sulphate contains sulphate ions.

Describe a test for sulphate ions.

test .....

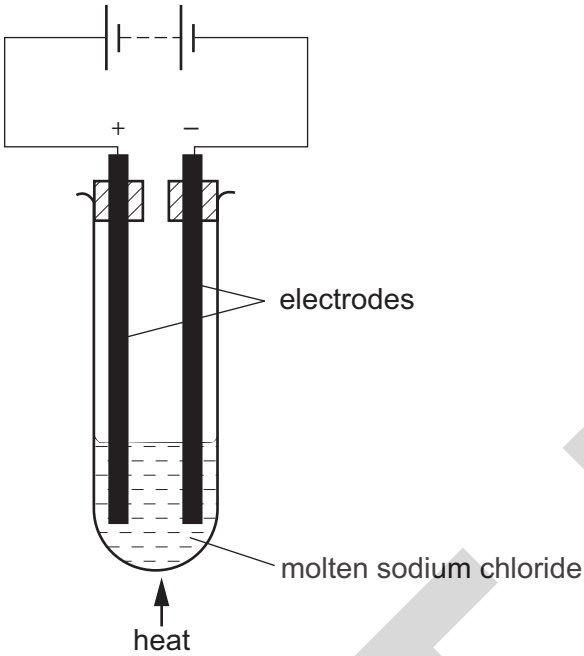
.....

result .....

..... [3]

For Examiner's Use

**(f)** Pure sodium chloride can be electrolysed using the apparatus shown below.




**(i)** Why does the sodium chloride have to be molten for electrolysis to occur?

.....

..... [2]

**(ii)** State the name of the product formed during electrolysis at

the anode (positive electrode) .....

the cathode (negative electrode) ..... [2]

**(iii)** Suggest a suitable substance which could be used for the electrodes.

..... [1]

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# DATA SHEET
## The Periodic Table of the Elements

| Group I | Group II | | | | | | | | | | | Group III | Group IV | Group V | Group VI | Group VII | Group 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 1 **H** Hydrogen 1 | | | | | | | | | | 4 **He** Helium 2 |
| 7 **Li** Lithium 3 | 9 **Be** Beryllium 4 | | | | | | | | | | | 11 **B** Boron 5 | 12 **C** Carbon 6 | 14 **N** Nitrogen 7 | 16 **O** Oxygen 8 | 19 **F** Fluorine 9 | 20 **Ne** Neon 10 |
| 23 **Na** Sodium 11 | 24 **Mg** Magnesium 12 | | | | | | | | | | | 27 **Al** Aluminium 13 | 28 **Si** Silicon 14 | 31 **P** Phosphorus 15 | 32 **S** Sulphur 16 | 35.5 **Cl** Chlorine 17 | 40 **Ar** Argon 18 |
| 39 **K** Potassium 19 | 40 **Ca** Calcium 20 | 45 **Sc** Scandium 21 | 48 **Ti** Titanium 22 | 51 **V** Vanadium 23 | 52 **Cr** Chromium 24 | 55 **Mn** Manganese 25 | 56 **Fe** Iron 26 | 59 **Co** Cobalt 27 | 59 **Ni** Nickel 28 | 64 **Cu** Copper 29 | 65 **Zn** Zinc 30 | 70 **Ga** Gallium 31 | 73 **Ge** Germanium 32 | 75 **As** Arsenic 33 | 79 **Se** Selenium 34 | 80 **Br** Bromine 35 | 84 **Kr** Krypton 36 |
| 85 **Rb** Rubidium 37 | 88 **Sr** Strontium 38 | 89 **Y** Yttrium 39 | 91 **Zr** Zirconium 40 | 93 **Nb** Niobium 41 | 96 **Mo** Molybdenum 42 | **Tc** Technetium 43 | 101 **Ru** Ruthenium 44 | 103 **Rh** Rhodium 45 | 106 **Pd** Palladium 46 | 108 **Ag** Silver 47 | 112 **Cd** Cadmium 48 | 115 **In** Indium 49 | 119 **Sn** Tin 50 | 122 **Sb** Antimony 51 | 128 **Te** Tellurium 52 | 127 **I** Iodine 53 | 131 **Xe** Xenon 54 |
| 133 **Cs** Caesium 55 | 137 **Ba** Barium 56 | 139 **La** Lanthanum 57 * | 178 **Hf** Hafnium 72 | 181 **Ta** Tantalum 73 | 184 **W** Tungsten 74 | 186 **Re** Rhenium 75 | 190 **Os** Osmium 76 | 192 **Ir** Iridium 77 | 195 **Pt** Platinum 78 | 197 **Au** Gold 79 | 201 **Hg** Mercury 80 | 204 **Tl** Thallium 81 | 207 **Pb** Lead 82 | 209 **Bi** Bismuth 83 | **Po** Polonium 84 | **At** Astatine 85 | **Rn** Radon 86 |
| **Fr** Francium 87 | 226 **Ra** Radium 88 | 227 **Ac** Actinium 89 | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 140 **Ce** Cerium 58 | 141 **Pr** Praseodymium 59 | 144 **Nd** Neodymium 60 | **Pm** Promethium 61 | 150 **Sm** Samarium 62 | 152 **Eu** Europium 63 | 157 **Gd** Gadolinium 64 | 159 **Tb** Terbium 65 | 162 **Dy** Dysprosium 66 | 165 **Ho** Holmium 67 | 167 **Er** Erbium 68 | 169 **Tm** Thulium 69 | 173 **Yb** Ytterbium 70 | 175 **Lu** Lutetium 71 |
| 232 **Th** Thorium 90 | **Pa** Protactinium 91 | 238 **U** Uranium 92 | **Np** Neptunium 93 | **Pu** Plutonium 94 | **Am** Americium 95 | **Cm** Curium 96 | **Bk** Berkelium 97 | **Cf** Californium 98 | **Es** Einsteinium 99 | **Fm** Fermium 100 | **Md** Mendelevium 101 | **No** Nobelium 102 | **Lr** Lawrencium 103 |

*58-71 Lanthanoid series
90-103 Actinoid series

Key

| a |
|---|
| **X** |
| b |

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).

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